

Project Proposal

Smart Family Task Manager with Location-Based Assistance

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Scope Document for Smart Family Task Manager

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1. Introduction:

The Smart Family Task Manager is a mobile-based application designed to help family members collaboratively manage daily tasks in an efficient and intelligent manner. In modern households, responsibilities are often shared among multiple members, but coordination becomes difficult due to busy routines and lack of awareness about each other's availability. Existing task management applications mainly focus on individual productivity and do not support real-time family collaboration. This project aims to introduce a location-aware task management system where tasks can be completed by any nearby family member. The application allows families to work as a single group, increasing cooperation and reducing delays in task completion. Location-based intelligence makes the system proactive rather than passive. Real-time notifications ensure transparency among all members. The system is especially useful for time-sensitive and location-dependent tasks. Overall, this project enhances family coordination using mobile and location technologies.

2. Problem Statement:

Families often face difficulties in managing shared responsibilities due to lack of coordination and real-time awareness. Tasks are usually assigned to a specific person even when another family member may be more conveniently located to complete it. Existing task manager applications do not support dynamic task takeover based on location. As a result, tasks such as bill payments, fee submissions, or grocery purchases are delayed unnecessarily. Family members are also not always informed when a task is completed, which causes confusion. There is no smart mechanism to suggest task completion based on proximity. Additionally, communication gaps lead to repeated efforts or missed responsibilities. Therefore, there is a need for a smart system that enables families to manage tasks collaboratively with real-time location awareness. The proposed system addresses these issues by allowing any nearby family member to complete pending tasks.

3. Problem Solution:

The proposed Smart Family Task Manager provides a centralized platform where all family members can add, view, and manage tasks collectively. Each task can be associated with a specific location, making it suitable for real-world activities. The system continuously monitors the location of family members with their consent. When a member comes close to a task location, the system automatically suggests task completion through notifications. Any member can complete the task and update its status. Upon completion, all family members are notified instantly. This approach improves efficiency and reduces dependency on a single individual. The system also maintains a task history for transparency. Overall, the application provides a smart, cooperative, and context-aware task management solution for families.

3.1. Objectives:

- BO-1: Improve family task completion efficiency through shared access.
- BO-2: Enable location-based task suggestions for nearby family members.
- BO-3: Reduce delays in completing time-sensitive tasks.
- BO-4: Increase transparency through real-time notifications.
- BO-5: Promote collaboration and shared responsibility within families.

4. Literature Review:

Application Name	Weakness	Proposed Project Solution
Cozi Family Organizer	No location-based task handling	Adds location-aware task suggestions
Google Tasks	Individual-focused task system	Provides family group-based tasks
Todoist	No real-time family collaboration	Enables shared family task completion

5. Scope:

The scope of this project includes the development of a mobile application that supports family-based task management. The system allows family members to create and share tasks within a group. Tasks can include optional location information to support location-based completion. The application provides real-time notifications to all members when tasks are created or completed. User authentication and family group creation are part of the system. Task status tracking and history logs are included. The application focuses on Android devices in the initial phase. Advanced AI features are not included in the current scope. Payment processing and third-party integrations are excluded. The system strictly supports family usage only.

6. Modules:

Module 1: User & Family Management

- FE-1: User registration and login
- FE-2: Family group creation and joining
- FE-3: Role assignment (Admin / Member)

Module 2: Task Management

- FE-1: Create, view, update, and delete tasks
- FE-2: Assign tasks to individuals or group
- FE-3: Task status tracking

Module 3: Location-Based Assistance

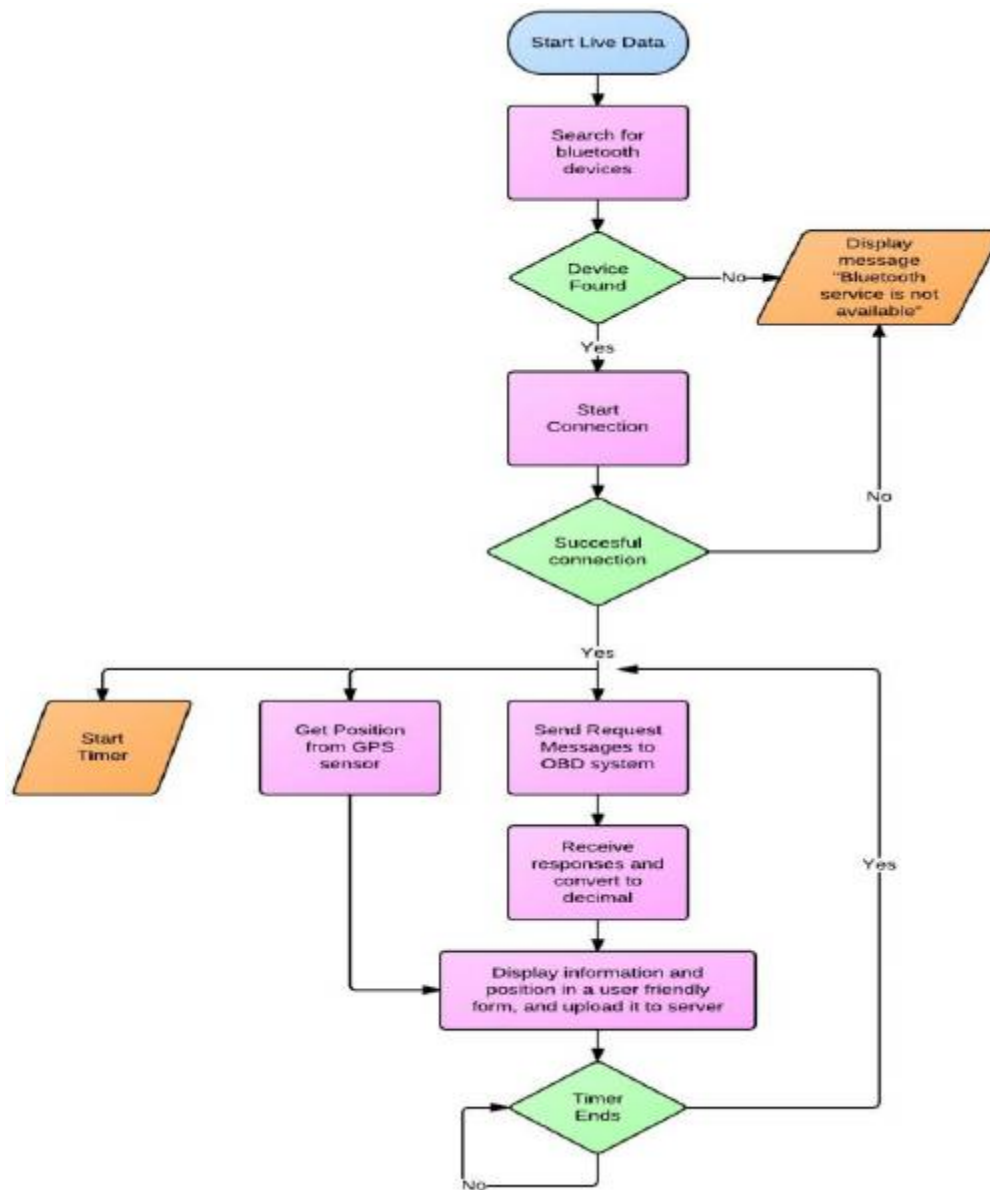
- FE-1: Attach location to tasks
- FE-2: Detect nearby family members
- FE-3: Suggest task completion

Module 4: Notifications & Logs

- FE-1: Real-time group notifications
- FE-2: Task completion alerts

- FE-3: Activity history

6.1. System Diagrams:



7. Constraints:

- LI-1: Requires active internet connection
- LI-2: Continuous GPS usage may affect battery
- LI-3: Location accuracy depends on device sensors
- LI-4: Privacy depends on user permissions

8. Data Gathering Approach:

The requirements were gathered through informal interviews with family members. Online surveys were used to understand common household task problems. Observation of daily family routines helped identify location-based tasks. Feedback was also collected from peers.

9. Tools and Technologies:

Tool / Technology	Version	Purpose
React Native	V19.2	Mobile App Development
Firebase	V15.1	Backend & Database
Android Studio	Otter 3	Development Environment

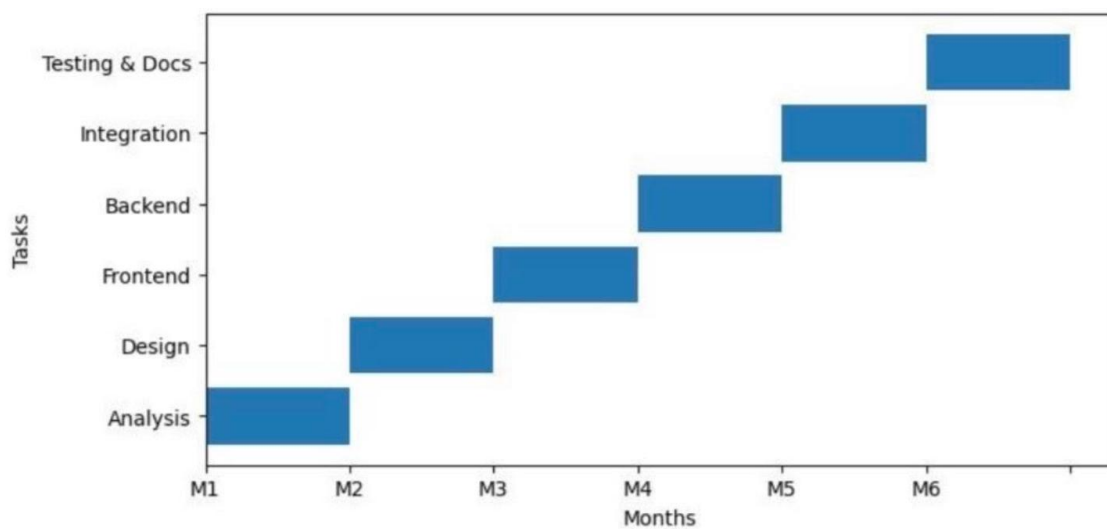
10. Project Stakeholders and Roles:

Stakeholder	Role
Air University Islamabad	Project Sponsor
Noor	Mobile App Development
Areeba	Backend & Database
Munaima	UI/UX & Documentation

Supervisor	Guidance & Evaluation
FYP Committee	Final Evaluation

11. Gantt Chart:

Month	Activities
Month 1	Problem identification, requirement gathering, literature review
Month 2	Scope finalization, proposal writing, system diagrams (ER & Use Case)
Month 3	UI/UX mockups, React Native learning, project setup
Month 4	Backend setup (Firebase, Auth, Database)
Month 5	Task Management + Location-Based Assistance module
Month 6	Notifications, testing, bug fixing, documentation & deployment



12. Mockups:

